Patrick Thangarajah

MECHANICAL DESIGN | TESTING & VALIDATION | PROTOTYPING | SIMULATION

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Summary

Recent MASc graduate in mechanical engineering with hands-on experience designing, building, and verifying hardware against requirements from concept through validation. Research background in occupant restraint crashworthiness, physical test correlation, DOE, and technical reporting.

Skills

Design & Prototyping SolidWorks, Onshape, Engineering Drawings, 3D Printing, Lathe, Mill, Bandsaw, Drill Press

Testing & Analysis DOE, JMP, Minitab, MATLAB, Microsoft Excel, Troubleshooting

Controls & Instrumentation Arduino, Sensors, Motors, Data Acquisition, Circuit Design, Altium, LTspice

Simulation & Programming FEA, LS-DYNA, LS-PrePost, Python, C/C++

Education

Master of Applied Science in Mechanical and Mechatronics Engineering *Waterloo, ON, Canada*

University of Waterloo *Sep 2023 - Jul 2026*

- Research in occupant restraint crashworthiness and injury biomechanics

Bachelor of Applied Science in Mechanical Engineering, Co-op Program *Vancouver, BC, Canada*

University of British Columbia *Sep 2018 - May 2023*

- Biomechanics and Medical Devices Option

Experience

Graduate Research Student - Impact Mechanics and Material Characterization Group *Waterloo, ON, Canada*

University of Waterloo *Sep 2023 - Jul 2026*

- Ran a 104-configuration LS-DYNA DOE and used JMP to quantify geometry effects on occupant response, identifying a configuration with 16% lower peak chest deflection
- Validated simulations against physical tests across two configurations and 10 response channels, achieving a 0.71 average ISO 18571 correlation score
- Troubleshot model-to-test discrepancies in positioning, restraint fit, and response timing to improve agreement with physical tests
- Automated seat belt fitting with LS-PrePost scripts, reducing repeated setup steps by 43% (7 to 4)
- Communicated findings through a master's thesis, conference paper, technical reports, and six sponsor presentations

R&D Engineer (Part-time) *Remote*

Echoscope *May 2023 - Sep 2023*

- Designed and iterated the bladder testing platform PCB, integrating pressure sensing, Arduino connections, power distribution, and motor and valve interfaces; refined component placement and routing and completed design-rule checks
- Authored five manufacturing and verification procedures covering PCB assembly, wiring, software setup, and functional testing; all five were approved and used by the client

Mechanical Engineering Intern *Airdrie, AB, Canada*

GN Corporation *Jan 2022 - Apr 2022*

- Developed standardized work instructions and SOPs by reviewing engineering drawings, customer specifications, and existing manufacturing processes
- Reduced fiberglass composite cutting waste by 50% through process analysis and proposed a material-handling workflow projected to reduce pick-and-place time by 50%

Engineering Intern *Victoria, BC, Canada*

Breathe Global *Jul 2021 - Aug 2021*

- Redesigned the analog control circuit for a negative-pressure breathing device in Altium, using LTspice simulation to select low-cost components
- Expanded the device's achievable breathing range from 70% to 88% of target

Projects

Effect of Seat Foam Properties on Injury Response in Rear Impact (ME 725 Course Project) *Waterloo, ON, Canada*

University of Waterloo *Jan 2024 - Apr 2024*

- Verified five polyurethane foam material models in LS-DYNA against input stress-strain response using single-element simulations, then implemented them in a rear-impact sled model with a Hybrid III ATD to calculate HIC15 and Nij injury metrics
- Determined that all five foams met the regulatory head injury limit but exceeded the regulatory neck injury limit at high speed, and that the softest foam reduced head injury risk by roughly 18–20% relative to the stiffest

Soft Distensible Bladder Model - Capstone Project *Vancouver, BC, Canada*

University of British Columbia *Sep 2022 - Apr 2023*

- Prototyped bladder phantoms using silicone and agar, comparing pressure-volume response with literature to guide material selection
- Designed and built the control and data-acquisition interface for a pressurized soft-bladder test system, enabling real-time pressure/volume monitoring and automated filling/voiding cycles
- Verified the system against defined performance requirements, demonstrating stable, repeatable operation up to 4.75 kPa internal pressure

Payload Team Member *Vancouver, BC, Canada*

UBC Unmanned Aircraft Systems (Student Design Team) *Sep 2019 - May 2021*

- Designed and prototyped a payload drop system and rover in Onshape and SolidWorks, iterating through 3D printed proof-of-concept builds
- Reduced payload system weight by 50% by repurposing components from previous designs during concept development

Conference Presentations and Publications

- **Thangarajah, P.** (2026). *Assessment of seat belt anchor position on rear seat occupant kinematics in frontal impact using a validated THOR-5F finite element model*. Master of Applied Science thesis, University of Waterloo, Waterloo, ON, Canada. hdl.handle.net/10012/23684
- **Thangarajah, P.**, Watson, B., Tang, K., Tylko, S., & Cronin, D. (2025). *Analysis of rear seat shoulder belt anchor position on kinematics of small-stature female occupants in frontal impact*. Injury Biomechanics Symposium, The Ohio State University, Columbus, OH, USA, May 18–21, 2025.